

To-Be-Fixed-3: Retrieval Latency Findings

Sarkari Sahayak — from eval run at alpha=0.6, candidate_k=15, rerank model ms-marco-MiniLM-L-12-v2

1. Rerank consistently costs ~1.1s on every query

Across all 10 eval queries, rerank_ms ranged narrowly from 1066ms to 1202ms with no outliers — this is a systemic, repeatable cost, not an anomaly. FlashRank reranking candidate_k=15 short candidates with ms-marco-MiniLM-L-12-v2 is adding roughly 1.1 seconds to every single voice turn, on top of embedding + search time. Fix options: switch to a smaller cross-encoder (e.g. ms-marco-TinyBERT-L-2-v2 instead of L-12), and/or reduce candidate_k from 15 to something like 8-10 so fewer candidates need to be scored per query.

2. embed_and_search had one 5.8s outlier (query q009)

9 of 10 queries had embed_and_search_ms between 444ms and 1008ms, but q009 ("assistance for farmers to sink a tubewell") spiked to 5799ms, pulling the mean up to 1220ms versus a median of 732ms. This looks like a one-off network hiccup with the Cloudflare embedding call rather than a systemic issue, but needs to be confirmed by re-running the eval to see if it reproduces on the same query. If it recurs specifically on that query, something is genuinely wrong; if not, treat the median (~732ms) as the representative embed+search cost rather than the mean.